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# Adventures Beyond Amdahl's Law: How Power-Performance Measurement and Modeling at Scale Drive Server and Supercomputer Design

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**Abstract** Amdahl's Law painted a bleak picture for large-scale computing. The implication was that parallelism was limited and therefore so was potential speedup. While Amdahl's contribution was seminal and important, it drove others vested in parallel processing to define more clearly why large-scale systems are critical to our future and how they fundamentally provide opportunities for speedup beyond Amdahl's predictions. In the early 2000s, much like Amdahl, we predicted dire consequences for large-scale systems due to power limits. While our early work was often dismissed, the implications were clear to some: power would ultimately limit performance. In this retrospective, we discuss how power-performance measurement and modeling at scale led to contributions that have driven server and supercomputer design for more than a decade. While the influence of these techniques is now indisputable, we discuss their connections, limits and additional research directions necessary to continue the performance gains our industry is accustomed to.

**Keywords** Amdahl's Law, speedup, power-aware computing, power modeling, performance modeling, performance prediction, power measurement

## 1 Introduction

In 1967, Gene Amdahl demonstrated, in a rather short paper[1], that parallelism has its limits. Simply put, for any given code, there is a fixed amount of parallelism available. Once this degree of parallelism (*DOP*) is exceeded by the available hardware parallelism typically related to the number of processing elements (*PE*), no additional speedup ($S$) is gained from further parallelism increases ($PE > DOP$).

In other words, the time it takes a code to complete ($T$) is the sum of the portion that is parallelizable ($T_p$) and the portion that runs only serially ($T_s$). Letting $T(1)$ be the time to run a given code on 1 PE and $T(N)$ the time the code takes to run on $N$ PEs, the speedup is expressed classically as:

$$\begin{aligned} S(N) &= T(1)/T(N), \\ S(N) &= (T_s + T_p)/(T_s + T_p/N), \end{aligned} \quad (1)$$

where $T(1) = T_s + T_p$. Thus, as $N \rightarrow \infty$,

$$S(N) = (T_s + T_p)/T_s. \quad (2)$$

This limitation has come to be known as Amdahl's Law and is still in use today—at times the concept is called strong scaling or fixed-workload speedup under an increasing number of PEs. The limits of strong scaling painted a dismal future for parallel computing researchers. In essence, Amdahl's Law discourages scaling the number of processing elements beyond a limit in the amount of parallelism inherent to a code.

The design of systems following Amdahl's declarations in some ways defied the conventional wisdom. For example, during the 1970's and 1980's the Cray 1[2] and later the Cray X-MP[3] respectively continued to increase system scale and enjoyed substantial commercial success while invigorating (or perhaps substantiating) the supercomputing industry.

Review

Special Issue in Honor of Professor Kai Hwang's 80th Birthday

Supercomputers were being used to solve the world's most challenging mathematical and natural science challenges while advancing the use of simulations of complex systems. This phenomenon was difficult to reconcile until a seminal paper by John Gustafson in 1988[4]. Gustafson observed that scientists using parallel programs increased their dataset sizes as parallelism increased. Smaller problems were run on smaller systems. Larger problems on these systems were impracticable but as larger systems became available, the larger problems could be run in reasonable amounts of time.

Gustafson further observed that Amdahl's predictions contained a critical assumption: namely, the workload for a parallel program was fixed in size ($W$) as the number of PEs increased. Gustafson drew a clear distinction between codes that used fixed workloads and those that scaled the workload for a single node ($w_1$) with the number of processing elements ($W = w_1 \times PE$). In practice, scientists would scale the workload as large as possible under some time limit. To capture this phenomenon, Gustafson proposed fixed-time speedup (in contrast to fixed-workload). Gustafson's Law explains why supercomputing and parallel processing have continued to benefit the scientific community. As long as problems and workloads grow, larger systems of processing elements will be needed.

These findings paved the way for additional speedup formulations. In 1993, Sun and Ni[5] demonstrated that for most codes, total system memory ($M$) was a key limitation on speedup. Memory-bounded speedup describes how parallel codes are limited not only by their DOP but also by their memory footprint. This idea can be applied to both fixed-workload and fixed-time scaling. When a code's total memory footprint ($MFP$) exceeds the total system memory bound ($MFP > PE \times (M/PE)$), speedup suffers.

## 2 Power of Parallel Processing

Each of the aforementioned speedup formulations played a part in a renaissance of supercomputer design. Parallelism is now implicit at every level of the system stack from microarchitecture pipelining to multicore and manycore processors, to general purpose graphical processing units (GPUs), to shared and distributed memory systems, to parallel servers, racks and full datacenters and system constellations and the applications that run on them. Parallelism transcends fundamental software in programming languages, compilers, operating systems, middleware, and tools and is heavily exploited in new concepts from virtual machines to containers to serverless computing.

These advances, often premised on parallelism, have led to an explosion of scale at all levels of the stack. For example, microarchitectures have closely followed Moore's Law for decades by doubling transistor density. Meanwhile, the insatiable need for performance drove the industry to couple more and more of these power hungry systems together. And while the supercomputing industry was focused on performance at all costs, our work[6–10] was the first to demonstrate that the speedup trajectory was not sustainable.

Today, everyone acknowledges that power is a challenge for large-scale systems. However, in 2002 when we first identified the coming power wall for supercomputers, hardly anyone noticed. In retrospect, this is not all that surprising. After all, legend has it that Seymour Cray once declared that Cray Research was primarily a refrigerator company given all the effort that went towards cooling their supercomputers. This is early evidence that while the supercomputing community is highly innovative, conventionally there was a conception that assembly of a large system composed of smaller commodity parts did not create new issues that could not be solved by the manufacturers.

Nonetheless, with significant effort, we assembled the first unassailable dataset of supercomputer performance and power data from a myriad of sources[6, 7]. The outcome was clear. Where historically, small rooms could provide enough power (tens of kilowatts) for a supercomputer, large rooms and buildings (hundreds of kilowatts) were increasingly common while systems were about to arrive exceeding multiple megawatts of power. In fact, the trajectory predicted power substations (tens of megawatts) and eventually power plants (hundreds of megawatts) were likely in the coming decade without drastic design changes.

## 3 Powerful Observations

With this data in hand, and against the prevailing wisdom in 2002, our research team created PowerPack[8, 11], a toolkit that combined the measurement of in-line-resistors using multimeters with software to align direct power measurements to system activity. Our work confirmed two hypothesis: 1) the energy consumption used in servers and supercomputers was significant, wasteful, and growing; and 2) the exploitation of inefficiencies common to parallel applications (e.g., busy waiting at barriers) could potentially reduce energy waste.

PowerPack provides power and energy measurement data of genuine import. The correlation of system activity with out-of-band power traces is essential to understanding the impact of software decisions on power consumption. Early on, this was intentional. Before the creation of the RAPL software APIs (Intel's Running Average Power Limit[12]) and other system in-band sensors for power tracking, PowerPack and its out-of-band, in-line system-level power profiles provided the only freely available, open-source method for identifying where power went and where the opportunities for savings could be found.

But, PowerPack goes one step further by separating the power consumption of individual devices within a server. This allows identification of the largest power producers in a system—typically CPUs, GPUs, memory, and mechanical disks (in this order). This also allows tracking of design changes over time—for example, early system measurements showed AC to DC conversion in cheap power supplies accounted for upwards of 30% total system power whereas newer systems have reduced this waste substantially. Another early indicator of future design improvements: CPU and memory accounted for as much as 75% of total system server power.

## 4 The Power of Influence

While we were somewhat confident that PowerPack would impact efficiency research in the years that followed, at the time we underestimated how the supercomputing and datacenter zeitgeists would catch fire in about 2008. It seems our dire predictions began to come true as municipalities started noting the regional demand for datacenter power and supercomputing centers began to build substations to support their machines. Large cities like New York began to cap the power to new datacenters and various whole industries (high-speed trading, e-commerce) suddenly looked threatened. This led to mainstream media attention and a burst of interest in power-performance data at scale[13].

For our part, we took several steps that in hindsight were of genuine influence. First, we inspired and supported dozens of research groups[14] worldwide with access to our PowerPack nodes, open source software, and expertise. This enabled dozens of researchers to explore their own hypothesis surrounding the impact of software and algorithm design changes on power in real systems. In particular, many sought to demonstrate ways to improve efficiency without sacrificing performance.

At the same time, we participated meaningfully in the design of SPECPower[15] and its suite of power-performance measurement and benchmarking toolkits which led to the Server Efficiency Rating Tool (SERT) framework[16] used for the EnergyStar Enterprise Server ratings. PowerPack's early experiences inspired and informed the design of these tools from the start.

PowerPack also provided a blueprint for the run rules of the Green500 List[17] of supercomputer energy efficiency. Since 2006, the Green500 List has pioneered the measurement, ranking, and tracking of energy efficiency in supercomputers. Major advances in server efficiency have been championed by the Green500 ranking and heavily influenced the adoption of systems using multicore, systems-on-chip, exotic memories, ARM-based processors, and the integrated use of accelerators. Now coupled with the Top500, the Green500 will likely continue to track new advances from application-specific systems designed for AI to the rise of quantum supercomputers.

## 5 Revisiting Amdahl's Law for Power-Aware Speedup

Power measurement has been essential to validating power-performance efficiency hypotheses and influencing design. However, it has also enabled us to uncover fundamental insights to the entangled nature of power and performance on real systems. In Kai Hwang's original and essential text on Parallel Computer Architecture[18], speedup using Amdahl's Law is defined traditionally as shown in (1) and (2). But, in the full equation, Prof. Hwang highlights two observations: 1) there are additional overheads for parallelism ($O_p$) that reduce effective speedup; and 2) these overheads are typically ignored for simplicity ($O_p = 0$) and because they are often either negligible or hard to measure. (2) then becomes:

$$S(N) = (T_s + T_p)/(T_s + T_p/N + O_p). \tag{3}$$

While building PowerPack, one of our early hypotheses was that Amdahl's Law would capture the effects of power management in much the same way that the law captured the diminishing returns of parallelism.

However, when we tried to model the effects of parallel system power management using (3), the conclusion was to always maximize the DOP until all code parallelism is exploited. This leads to maximizing the power use at the scale of $PE = DOP$. However, in practice, we found the most efficient settings were rarely $PE = DOP$ as there were opportunities to maintain a high DOP while throttling down the systems at certain times to improve efficiency. For one, the ability to save energy without impacting performance is tied closely with the compute to communication ratio. When a code is dominated by compute, running things fast on a processor leads to high efficiencies since the code finishes more quickly. But, when memory and communication time dominate, there are more opportunities to slow the processor (reduce power) without losing performance.

Amdahl's Law does not capture these effects as they are tied to the split between time spent executing computations and time spent waiting on memory or communications. So we explored whether other speedup laws (fixed-time, memory-bounded) would capture these effects. But, our work showed that all of the existing laws focused on degree of parallelism and changes in total workload, not distinguishing between compute time and communication time precisely.

A key observation early on was that reducing the power to CPUs was most effective during long intervals of communication. For example, as CPUs block on a barrier in parallel, spinlocks are simple and fast but waste energy significantly. Prediction models[6, 10] that guess when CPUs spin on communication and for how long provide opportunities to reduce CPU energy waste. The more the communication that takes place, the more the energy savings recovery without performance loss. But, as stated above, speedup formulae typically ignore communication overhead—precisely where the energy savings are possible.

With this fundamental observation, we realized that parallel speedup models for power had to consider communications time (effectively the $O_p$ term in (3)) to motivate the use of efficiency techniques. The resulting model, Power-aware Speedup[19], showed conclusively that Amdahl's Law could be generalized to consider both computational workload and communication overhead. By separating the workload, the effects of power management can be considered separately—where there are minimal effects on overall speedup when power management is engaged and communication overhead dominates. This allowed us to accurately model speedup as codes scale under power-aware constraints.

## 6 Iso-Energy Efficiency and Power-Capping

Power-aware speedup led to a deeper understanding of the role of power management in parallel systems under performance constraints and resulted in additional insights. Power-aware speedup provides a way to couple changes in workload and power consumption to performance in parallel and distributed systems. In essence, power-aware speedup empowers us to view each of these parameters (voltage and frequency, workload, the number of threads) as control knobs[①] we can adjust depending on our target goals (power, performance, efficiency).

In the original application of power-aware speedup, we attempted to isolate the amount of time spent in communication to exploit the overhead for improvements in energy efficiency by scaling down processor speeds for fixed-size workloads. But, the fundamental insights are more broadly applicable. For example, we can fix efficiency (say flops per watt) and control the system "knobs" (i.e., workload, thread count, processor speed) for a target efficiency. In some ways, this is akin to iso-efficiency[20] or fixed-performance scaling. Thus, we named this type of system steering "iso-energy-efficiency"[21] where we manipulate these knobs at runtime to maintain a fixed efficiency.

Power-aware speedup also encourages fixing the power and maximizing the performance. Moreover, the iso-energy-efficiency techniques can be used to identify a target efficiency under a fixed power (e.g., a power cap) and tune the previously mentioned knobs to maximize performance.

## 7 Power-Performance Overlap

The aforementioned techniques focus on the use of one or two power managed devices simultaneously. Early systems focused on throttling of CPUs, thread

①In engineering, a control knob (or knob) is a rotary device used to provide manual input adjustments to a mechanical or electrical system when grasped and turned by a human operator. Here we invoke this imagery to denote human-driven software or hardware parameter control. While our focus is on power-performance control, knobs are present throughout computer systems as settings (e.g., page size), limits (e.g., page buffers), protocols (e.g., write-through caches, FCFS scheduling), etc.

counts, etc. However, technological advances have led to more, varied, power-controllable devices including GPUs and DRAM. The techniques proposed in the early days of high-performance power management fail or operate inefficiently when one or two independent components attempt to save power and act simultaneously in isolation.

Initially, we struggled to identify adaptation of our early techniques for these more complicated, siloed power managed systems. While some simulations showed promise[22], we found that on real systems when CPU, memory, and threads were managed the existing techniques performed poorly.

A key insight came when we considered the Power-aware Speedup of these techniques. Effectively, implementing Power-Aware Speedup concepts taught us to break down the workload into workload affected and unaffected by throttling. Applying this thinking to CPU, memory, and threads meant identifying which pieces of the workload were affected by CPU speed, memory speed, and thread changes as well as their pairwise products (CPU + memory, memory + threads, CPU + threads).

This insight led us to identifying computational overlap ($O$) as a proxy for the separation of the workload into three pieces (Compute ($C$), Overlap ($O$), and Stalltime ($S$)). The original application of power-aware speedup effectively coupled the overlap of memory and communication with either the computation or the stalltime. This insight explains why all prior techniques, from our work and many others[23–25], were ineffective when these three throttling techniques were used simultaneously. COS[26] is a more precise modelling instrument for power-scalable systems that captures the nuanced power-performance relationships that result from complex memory hierarchies, non-blocking caches, out-of-order execution, parallel communications, etc.

## 8 The Power and Potential of Emergent Technologies

In the time since our early publications[6, 7], interest in and exploration of power-aware, high-performance computing has exploded with energy efficiency a key design constraint in all systems of scale: servers, datacenters, supercomputers, and clouds. Whole industries have adapted to this new paradigm thanks to the follow-on and continued efforts of dozens of research groups and thousands of researchers over the past two decades.

Currently, we live in the age of power capping where large-scale systems are designed to fit within a power envelope. This has not removed the need for energy efficiency—to the contrary, now it is more important than ever for every device and the entire system to run efficiently, maximize performance, and keep power costs manageable in service to scientific discovery.

The future is ripe for transformational research. Quantum computers are emerging and it is likely that technologists will find uses for them to scale applications in ways previously impossible. These systems, or at least the designs at present, will likely consume inordinate amounts of power for cooling while creating conditions for reliable quantum behaviors.

Quantum systems are effectively limited in their applicability to certain types of problems. In essence, based on current designs, quantum computers will not be general purpose and will likely be applicable to a class of problems akin to the way we exploit other accelerators (GPGPUs, FPGAs) today. This means quantum systems will augment rather than replace conventional designs of large groups of tightly coupled off-the-shelf systems.

## 9 Conclusions

The ground-breaking revelations of Amdahl's Law and our own predictions of power use in large-scale systems shared a single premise: namely, these projections dictate our future only if we do not adapt. Luckily, the past 50 years of Amdahl's Law and 20 years of Green HPC have demonstrated the power of innovation to avoid dire consequences for parallel and distributed computing.

In any case, speedup and performance models must be adapted to continue to identify the key performance attributes of emergent systems. These models give us the notion of ideal performance, power, and efficiency as well as the ability to identify weaknesses in current designs. The application of these models and techniques motivates and informs the design of future systems (SPECPower, Green500) of scale. For example, the development of the Intel RAPL (Running Average Power Limit)[27] "sensors" uses performance models akin to those initially developed in PowerPack to estimate power consumption for memory. More measurements throughout future systems made available to power-performance engineers and system developers will enable new inputs to future models that improve efficiency. Analogously, as system devices offer more power-performance states with corresponding control knobs, there will be additional opportunities to create models that offer new insights and runtime efficiency steering.

No matter which path we take, the future is bright, exciting, and wrought with fundamental challenges that will continuously threaten our ability to deliver the highest performance under power constraints. While the destination is admirable, we believe innovation awaits us at every turn in the journey.

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